Hardik Anil Kolhe

Data Scientist | AI/ML Engineer | Building Production AI Systems with LLMs, RAG & Predictive ML

Aim to leverage machine learning, AI, and advanced analytics expertise to build production-grade predictive models, end-to-end data solutions, and AI-driven products that drive measurable business, operational, and engineering impact.

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PROFILE SUMMARY

AI creates value only when it solves real business problems. I am a Data Scientist and AI/ML Engineer with experience building production-ready Machine Learning and Generative AI systems across renewable energy, healthcare, e-commerce, and enterprise analytics. My expertise includes predictive machine learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), MLOps, cloud deployment, and scalable AI pipelines. Enjoy transforming complex data into intelligent products that deliver measurable business impact.

AREAS OF EXPERTISE

- Machine Learning & Predictive Analytics
- Generative AI, LLMs & RAG Applications
- End-to-End ML Pipeline Development
- MLOps, Production Deployment & CI/CD
- Data Engineering & Model Optimization
- Time Series Forecasting & Predictive Maintenance
- Feature Engineering & Advanced Data Preprocessing
- Statistical Modeling & Analytical Problem Solving
- Scalable AI Application Development & Deployment
- Explainable AI (SHAP/LIME)
- Data Visualization & Business Intelligence
- Collaboration & Stakeholder Communication

TECHNICAL SKILLS

- **Machine Learning & Deep Learning:** TensorFlow, Scikit-learn, XGBoost, CNNs, Transformers, Gradient Boosting, Ensemble Models, Time Series Models
- **Generative AI & LLM Systems:** RAG, OpenAI API, LangChain, FAISS, Prompt Engineering, Vector Embeddings, Agentic AI
- **ML Engineering & Deployment / MLOps:** Python, FastAPI, Django, Flask, Streamlit, REST APIs, Docker, Airflow, CI/CD, PyInstaller, Model Monitoring & Retraining Pipelines
- **Data Processing & Analytics:** Pandas, NumPy, SciPy, Feature Engineering, Data Cleaning, EDA, Large-Scale Data Handling, Customer Segmentation, RFM Analysis, K-Means Clustering, KNN
- **Cloud, Databases & Visualisation:** AWS, GCP, BigQuery, PostgreSQL, Power BI, Matplotlib, Seaborn, Plotly, Dashboard Development, Google Analytics 4 (GA4), Shopify Analytics, Stripe
- **Tools & Workflow:** Linux, GitHub, Jupyter, VS Code, Containerised Workflow

WORK EXPERIENCE

Data Scientist | Sriya.AI , Georgia, United States — July 2026 – Present

- **AI Product Development:** Developed production-ready AI solutions for the Sriya AI platform, delivering **Website Audit, Customer Intelligence, and Google Analytics 4-**driven business insights through scalable machine learning workflows.
- **Machine Learning Engineering:** Built end-to-end ML pipelines covering data ingestion, preprocessing, **feature engineering**, predictive modelling, and **automated analytics**, enabling intelligent decision support for enterprise applications.
- **Generative AI & LLM Systems:** Designed and integrated **Large Language Models (LLMs), BigQuery integrations**, and AI-powered reporting workflows to generate automated business insights, recommendations, and executive reports.
- **Backend & Cloud Integration:** Implemented secure backend services using Django, REST APIs, **PostgreSQL**, Stripe, **OAuth 2.0**, and **Google Analytics APIs**, supporting licensing, subscriptions, and payment workflows.
- **Collaboration & MLOps:** Worked closely with engineering and product teams to deliver scalable **AI applications** while applying CI/CD and **MLOps** best practices for deployment, monitoring, and continuous improvement.

Data Scientist Intern | Sriya.AI , Georgia, United States — May 2026 – July 2026

- **AI Application Development:** Developed AI-powered Website Audit and **Google Analytics 4 (GA4)** analysis workflows using Python, Django, **OpenAI**, and **Google Analytics APIs**, automating website analysis and business insight generation.
- **Customer Analytics:** Built machine learning models using **RFM Analysis,** K-Means Clustering, KNN, and **customer segmentation** techniques to support marketing optimization and customer intelligence.
- **Data Engineering:** Developed production-ready **Master Datasets** through data cleaning, feature engineering, missing value imputation, and target variable creation for scalable machine learning pipelines.
- **Generative AI Solutions:** Designed **LLM**-based workflows for automated report generation, intelligent business recommendations, and enterprise analytics using **OpenAI** and **Agentic AI** concepts.
- **Deployment & Integration:** Assisted in desktop application deployment using PyInstaller, **PostgreSQL**, SQLite, Stripe, and **REST APIs**, contributing to the commercial release of the Sriya AI platform.

Machine Learning Engineer Intern | Suzlon Energy Limited, Hengelo, Netherlands March 2025 – September 2025

- **Hybrid Predictive Modelilng:** Engineered production-grade pipelines using XGBoost and **Physics-Informed Neural Networks (PINNs)** to forecast wind turbine fatigue life with **73% accuracy**, ensuring predictions adhered to mechanical engineering principles.
- **Model Optimization & Interpretability:** Streamlined model complexity by reducing features from **15 to 8** using SHAP-based interpretability and automated hyperparameter tuning via **Optuna**, significantly accelerating model iteration cycles.
- **MLOps & CI/CD Deployment:** Architected end-to-end deployment workflows using **Docker and GitHub Actions**, implementing automated retraining and monitoring to maintain model performance within a cloud-based R&D infrastructure.
- **Data Engineering & Anomaly Detection:** Developed a robust preprocessing framework for heterogeneous datasets, implementing log-transformations and **anomaly detection** on sensor data to flag early warning signals and reduce unplanned downtime.
- **End-to-End Analytics & Reporting:** Created automated dashboards and reporting tools to visualize fatigue predictions, transforming complex model outputs into actionable maintenance schedules for technical stakeholders.

EDUCATION

- **Master of Science in Data Science and Analytics |** EPITA University, Paris, France | Sept 2023 – Jan 2026
  Specializsd in Data Science, Artificial Intelligence, Machine Learning, and Generative AI.
  Thesis: Developed a CNN-based heart stroke prediction model using multimodal medical data.

- **Bachelor of Engineering in Information Technology |** University of Mumbai, Mumbai, India | Aug 2018 – May 2022
  Built a strong foundation in software engineering, databases, machine learning, and data analytics.

Refer to the annexure for project details

ACADEMIC PROJECTS ANNEXURE

Heart Stroke Prediction (AI Diagnostic Tool)
- Developed a hybrid CNN + Gradient Boosting model on ECG leads V5 & V6 and clinical patient data, achieving 95%+ accuracy.
- Built a HIPAA-compliant clinical interface with doctor and patient portals using Streamlit.
- Integrated real-time ECG and patient record processing via AWS cloud services.
- Enhanced model robustness using image preprocessing, feature selection, and data balancing techniques.
- Enabled early detection of cardiac events, reducing treatment delays and healthcare costs.

Olympic Games Data Analysis & Visualisation (1896–2022)
- Analysed 314K+ records spanning 126 years to uncover hosting trends, medal distributions, and national performance patterns.
- Cleaned and standardised datasets to handle missing values and ensure analytical accuracy.
- Created interactive geospatial dashboards and choropleth maps using Plotly and GeoPandas.
- Highlighted insights such as "home advantage" (host nations winning 15–25% more medals) and global hosting disparities.
- Delivered actionable insights on performance evolution, national strengths, and sport development strategies.

Project Report Analyser (RAG Application)
- Built a Retrieval-Augmented Generation (RAG) system for intelligent Q&A across multiple industrial PDF reports.
- Developed scalable PDF ingestion and chunking pipeline with structured metadata (document name, page number) for precise semantic retrieval.
- Implemented OpenAI embeddings with FAISS vector database to enable cross-document semantic search and multi-project comparative analysis.
- Designed a context-constrained prompting framework ensuring grounded, citation-backed responses.
- Created an interactive Streamlit interface supporting multi-file uploads, natural language queries, and structured outputs.
- Engineered fallback logic for out-of-scope queries and optimized retrieval performance for enterprise-level deployment.

Car Price Prediction Web Application & Monitoring System
- Built an end-to-end ML system with FastAPI backend, Streamlit frontend, and PostgreSQL database.
- Developed automated ETL, model retraining, and validation pipelines using Airflow.
- Incorporated Great Expectations for data quality validation and implemented Grafana dashboards for monitoring model health and performance metrics.
- Optimized batch processing with efficient chunking and database query tuning.
- Delivered a modular, scalable architecture adaptable for diverse ML prediction use cases.